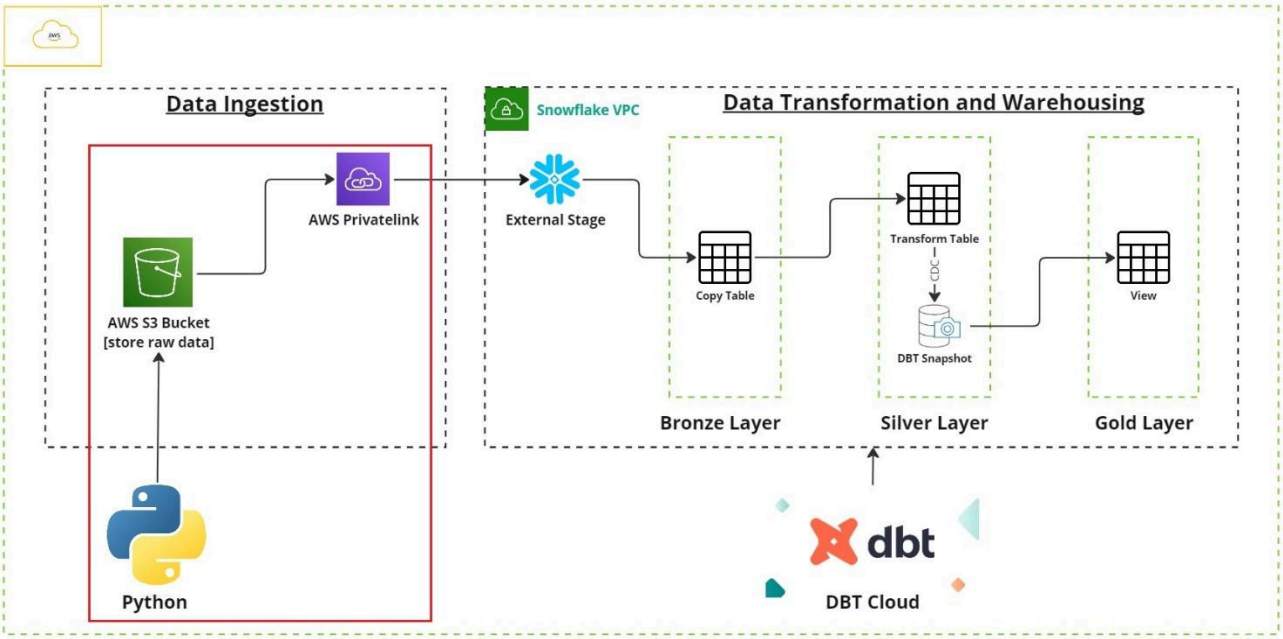
Technical Overview

We are going to setup AWS S3 bucket. Also, will load the '.csv' data file into the bucket using a python script. This will ensure we have our data loading capability from our local systems.

Architecture



Pre-requisites:

- **AWS Account**

Configuring AWS CLI

If you have not already done so you will need to configure AWS CLI to your account.

Windows:

Install AWS CLI. Use link - https://docs.aws.amazon.com/cli/v1/userguide/cli-chap-install.html

Choose your Windows operating system and follow the instructions on the page.

- **Download the appropriate MSI installer**
- **Run the downloaded MSI installer or the setup file.**
- **To confirm the installation, use the aws --version command at a command prompt (open the Start menu and search for cmd to start a command prompt).**

MacOS:

Install AWS CLI. Use below commands to install the latest version:

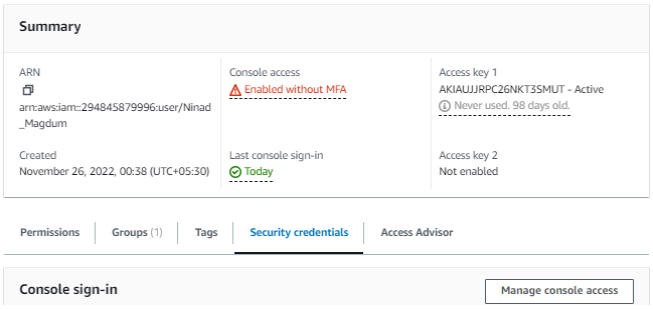
- **curl "https://s3.amazonaws.com/aws-cli/awscli-bundle.zip" -o "awscli-bundle.zip"**

unzip awscli-bundle.zip

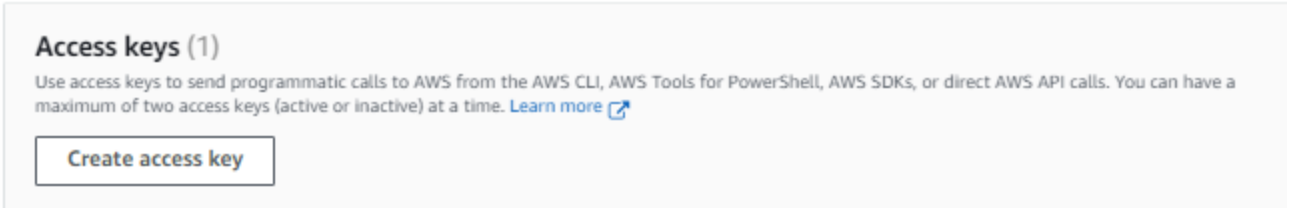
sudo ./awscli-bundle/install -i /usr/local/aws -b /usr/local/bin/aws

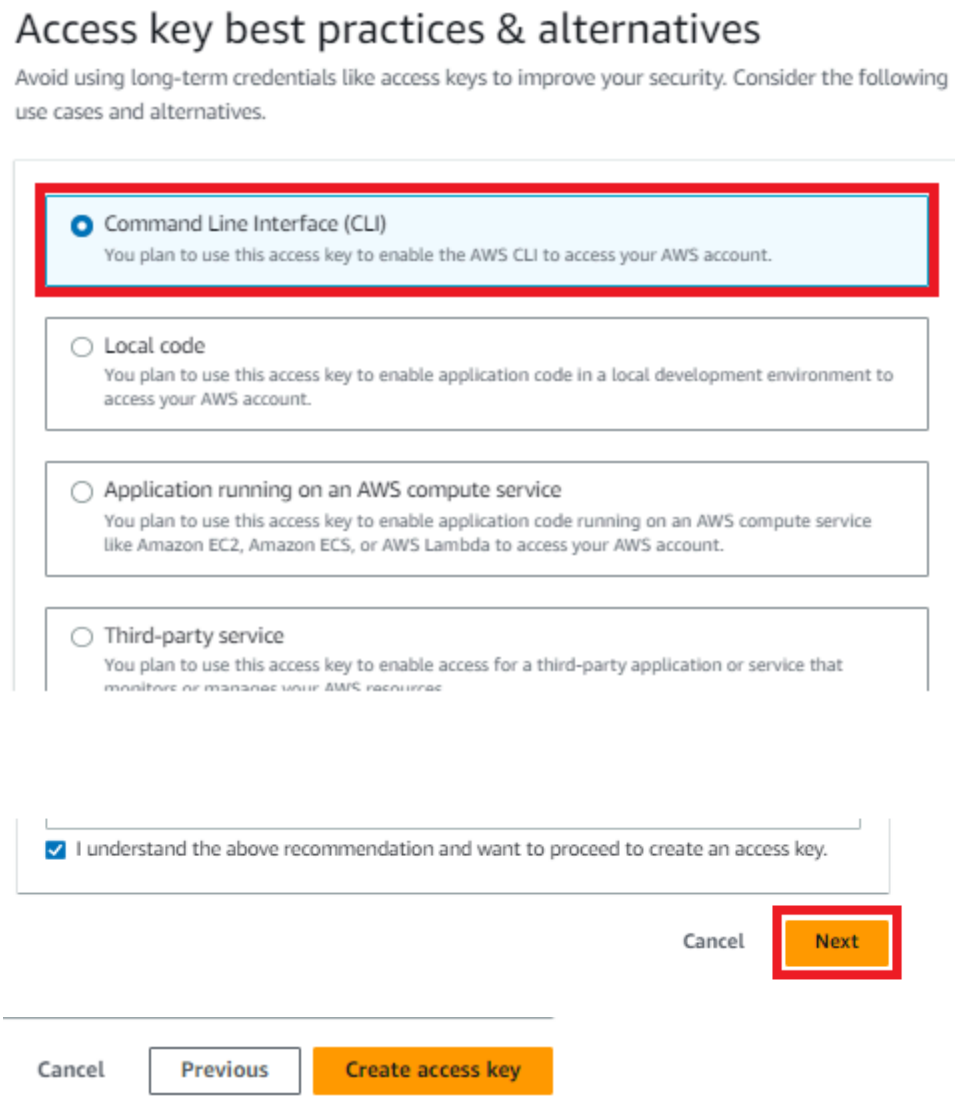
1. Navigate to the user’s section of the IAM dashboard.

2. Select the user you are logged into.(check top right corner for your username) Click the "Security credentials" tab.

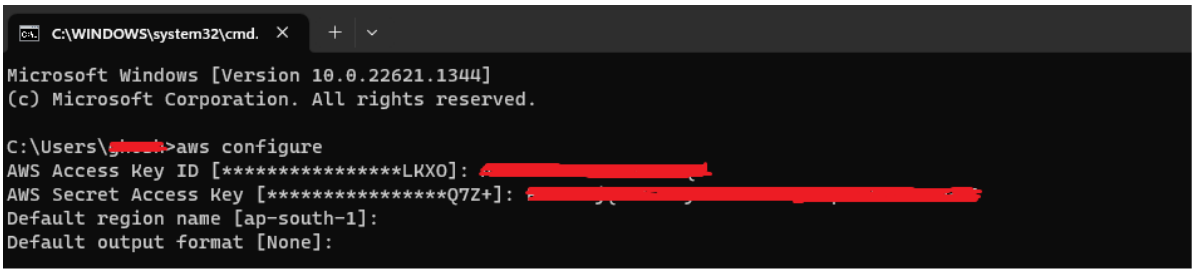


3. Click "Create access key". Copy down both the Access key ID and secret access key to a secure location.



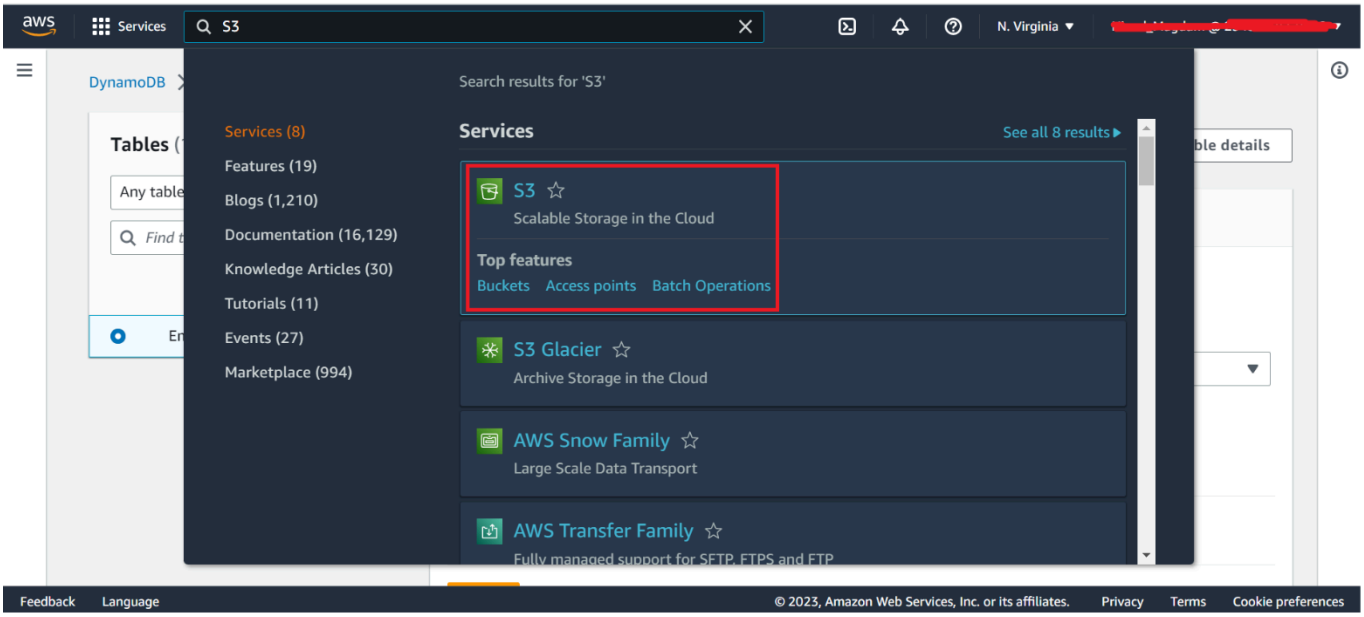


4. Open a terminal on your local machine and enter ‘aws configure’. Enter the access key and secret access key from Step 4. Choose a default region name and output format, the defaults are fine if you are not sure.

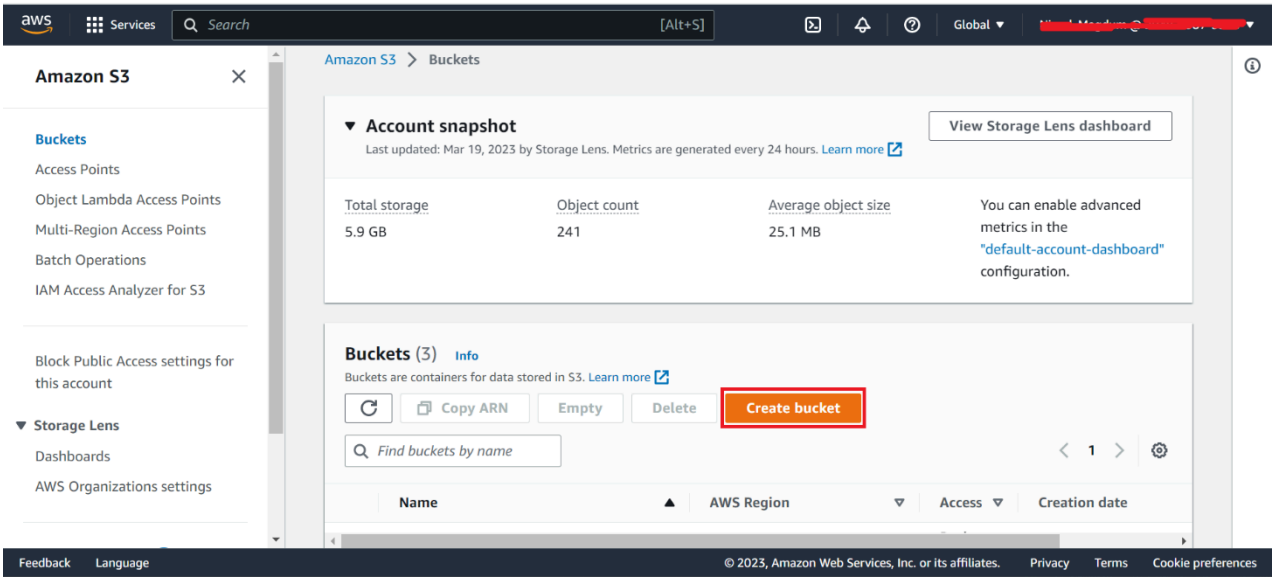


S3 Bucket Setup

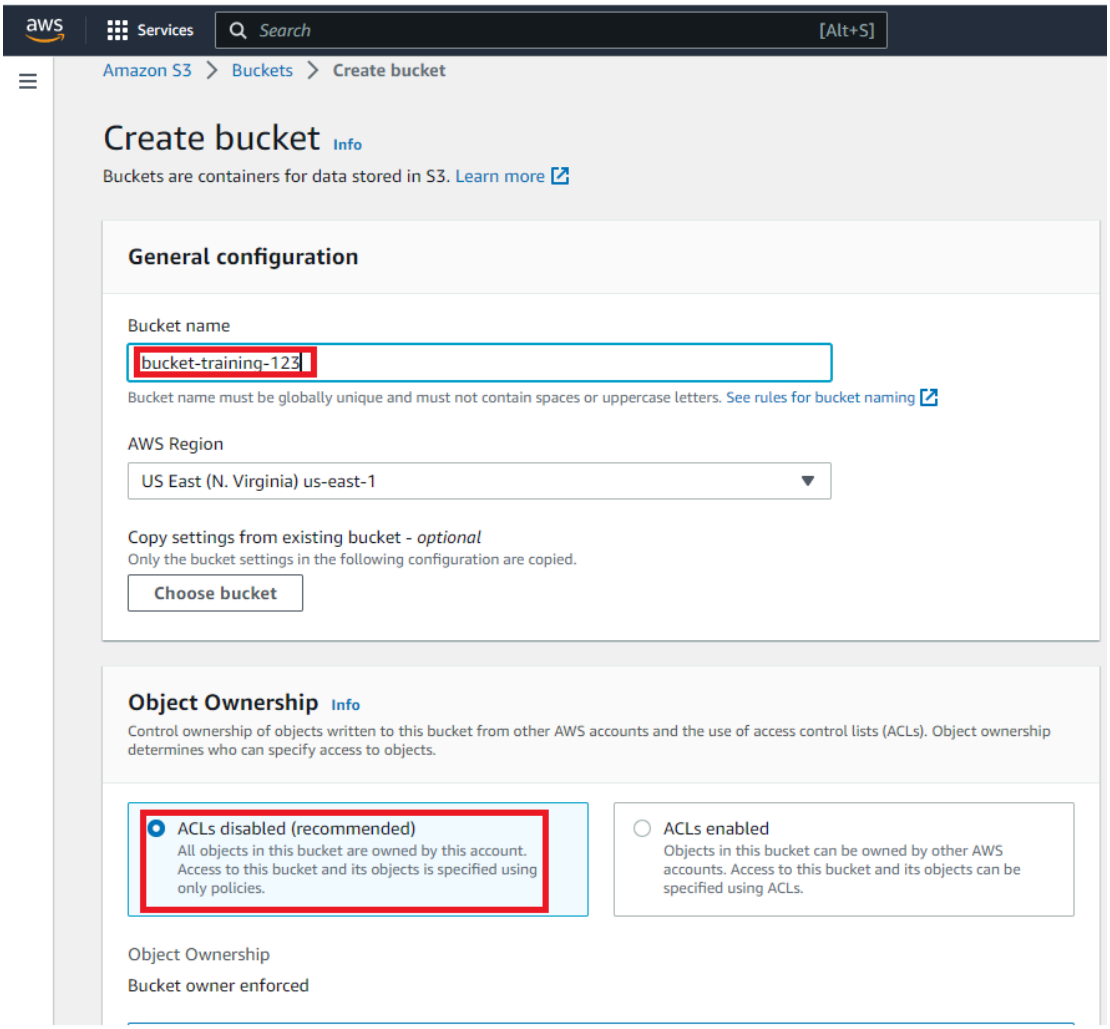
1. Go to AWS console and search for S3. Select it from the result list.

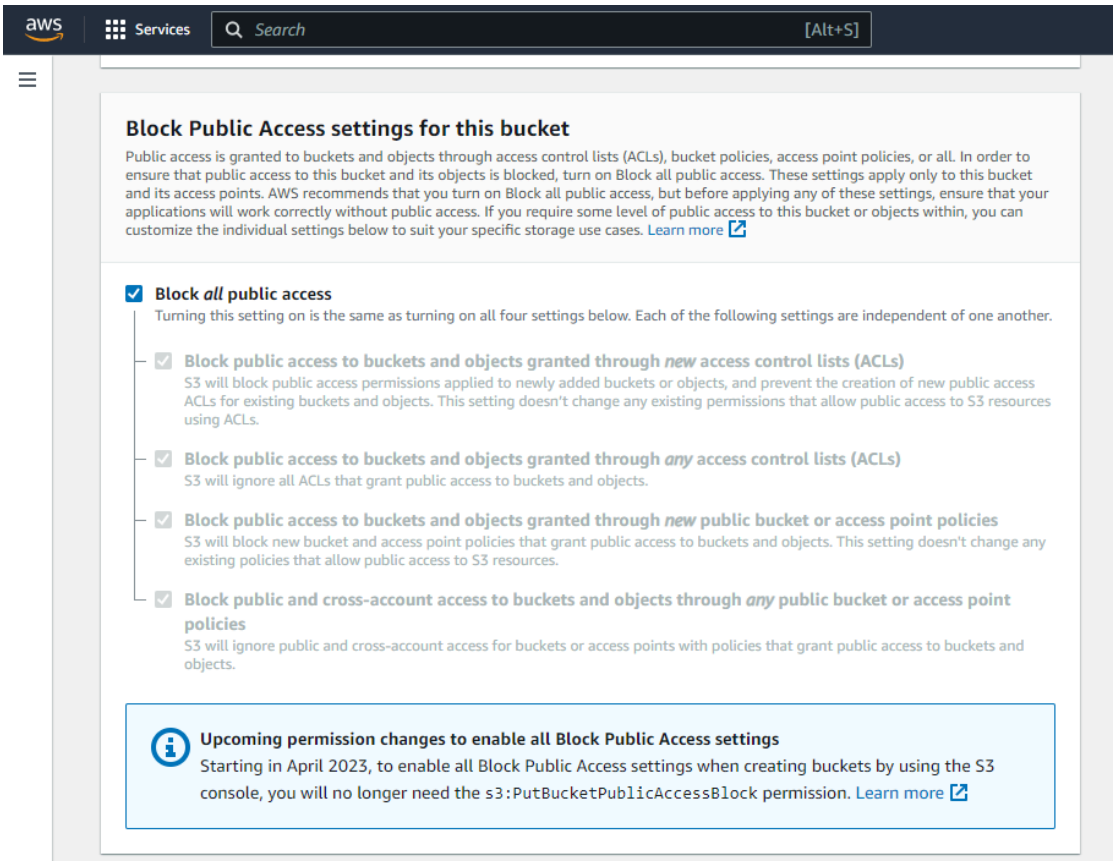


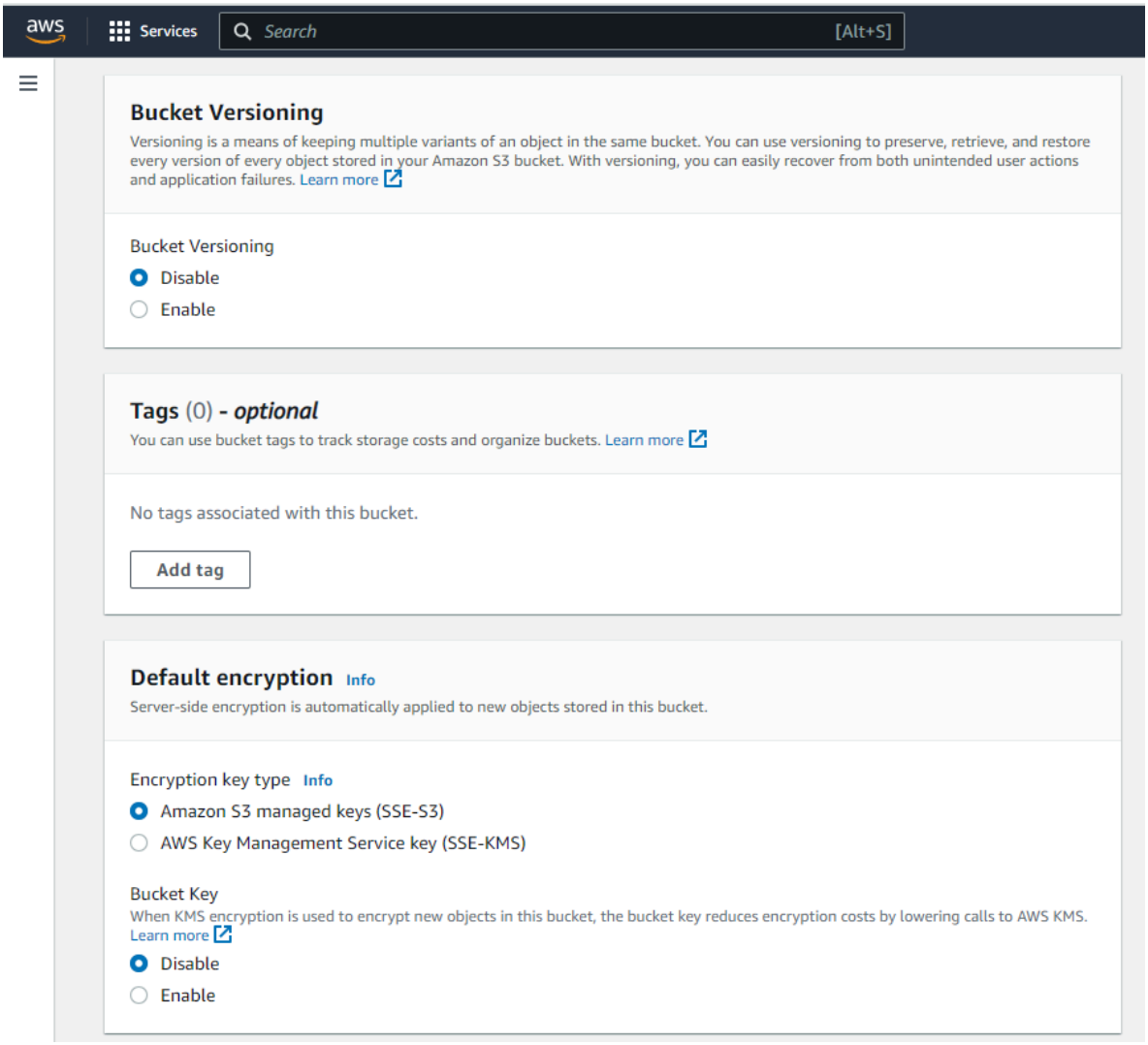
2. Then on the next page click on Create bucket.



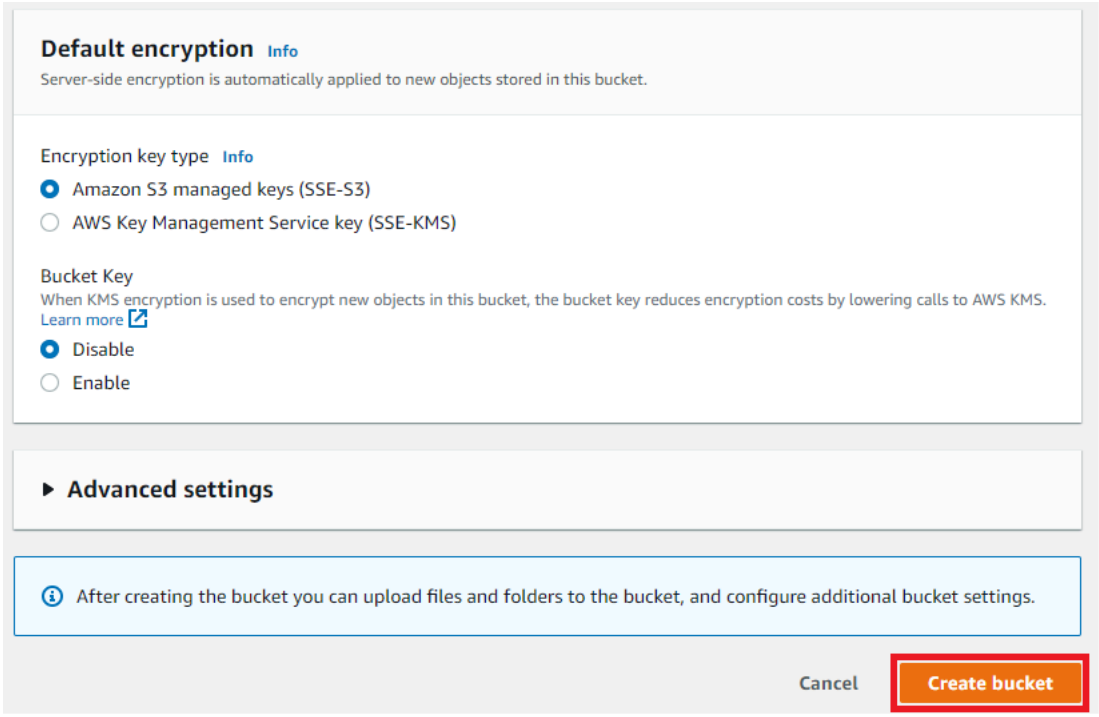
3. Now add a name for your bucket and keep settings as shown.



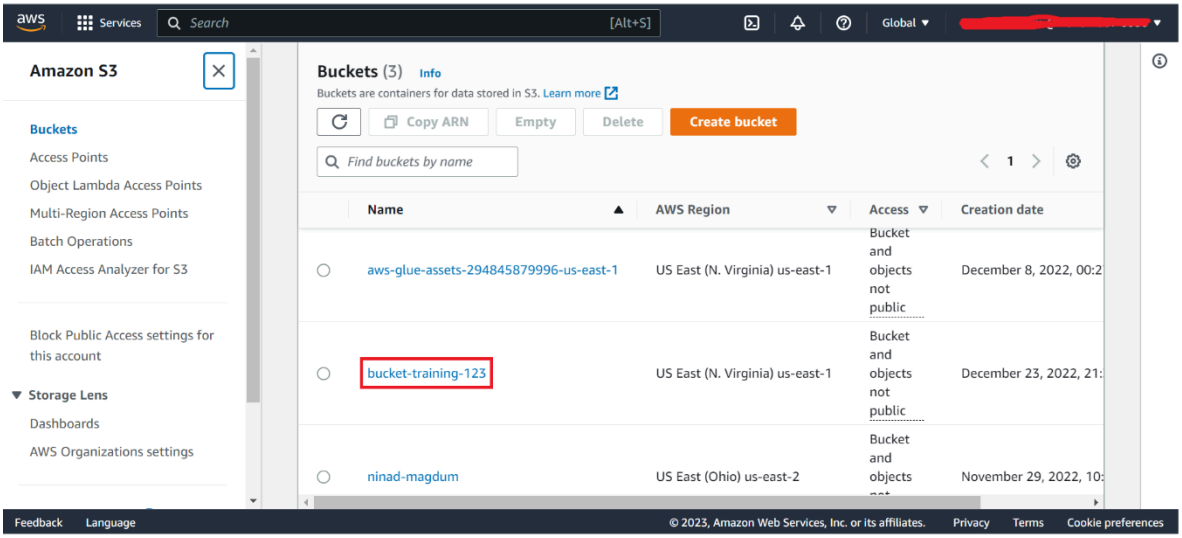

aws
Services
Search
[Alt+S]
Block Public Access settings for this bucket
Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. Learn more
Block all public access
Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.
Block public access to buckets and objects granted through new access control lists (ACLs)
S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.
Block public access to buckets and objects granted through any access control lists (ACLs)
S3 will ignore all ACLs that grant public access to buckets and objects.
Block public access to buckets and objects granted through new public bucket or access point policies
S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.
Block public and cross-account access to buckets and objects through any public bucket or access point policies
S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.
Upcoming permission changes to enable all Block Public Access settings
Starting in April 2023, to enable all Block Public Access settings when creating buckets by using the S3 console, you will no longer need the s3:PutBucketPublicAccessBlock permission. Learn more



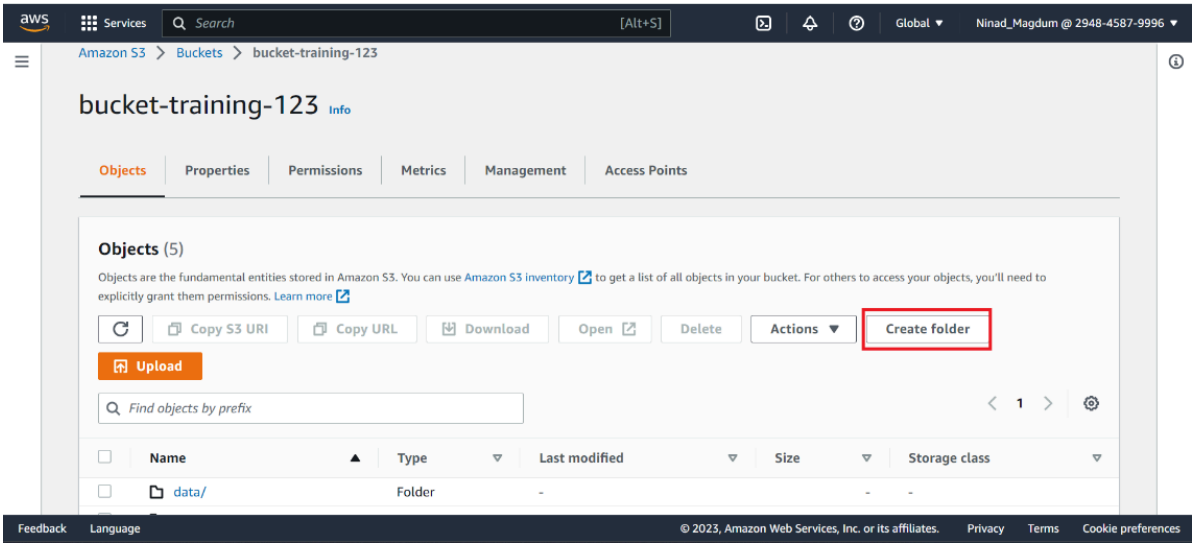
4. Then click on Create bucket to proceed. Your bucket has been created.



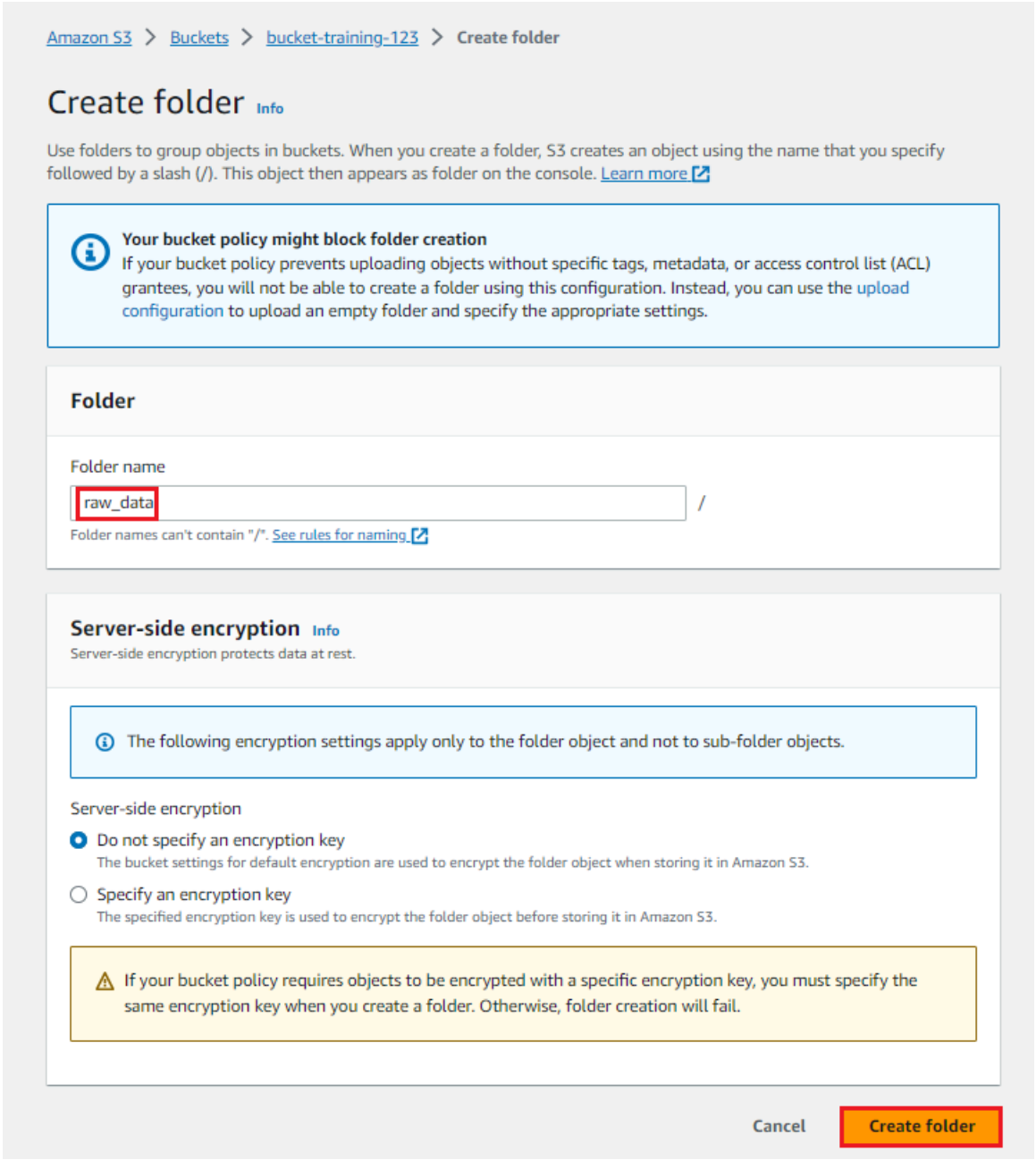
5. Now find your bucket from the list and click on it.



6. Then click on Create folder.



7. Now fill in the folder name details like raw_data and click on Create folder.



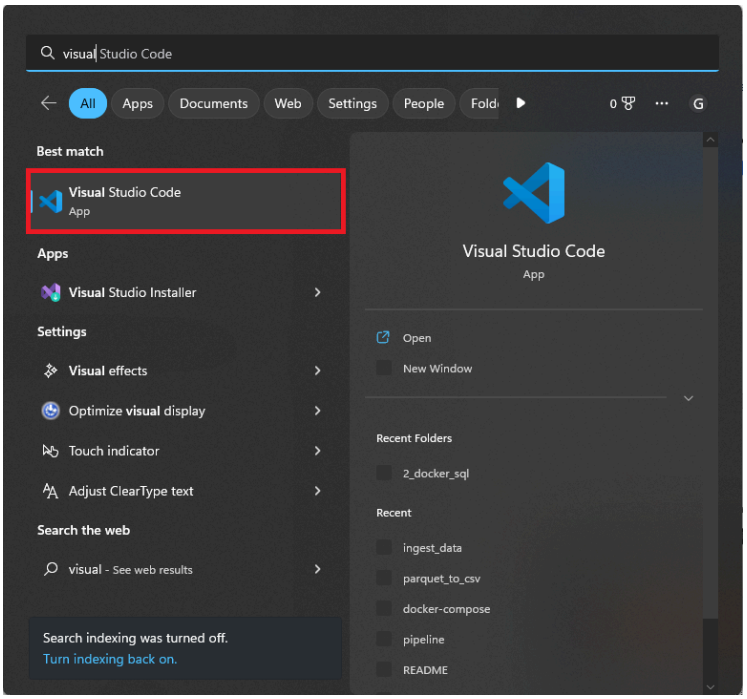
Loading data from Local

1. Loading data to AWS S3 bucket folder using a .csv file can be done using python. For this project we will use product data file. The dataset has been provided with this course document.

https://drive.google.com/drive/folders/18p-NQY7n12tmJIjrVK46Cuy97Gt39HA_?usp=sharing

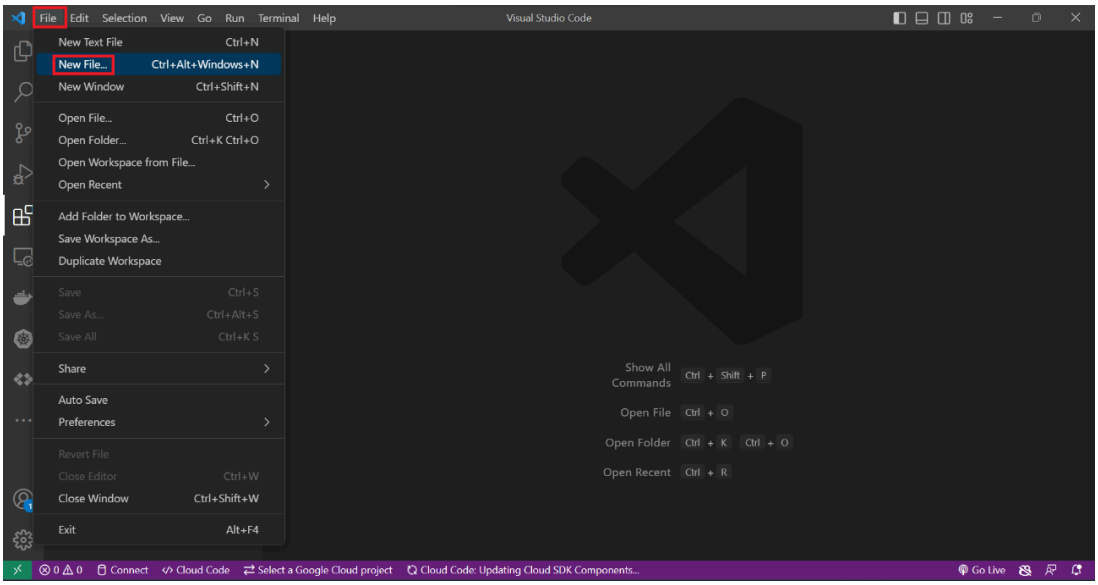
2. Download and Install visual studio. Use link for Windows & MacOS: https://code.visualstudio.com/download . Choose your required operating system and proceed. Once downloaded run that file to install.

3. Run Visual Studio Code

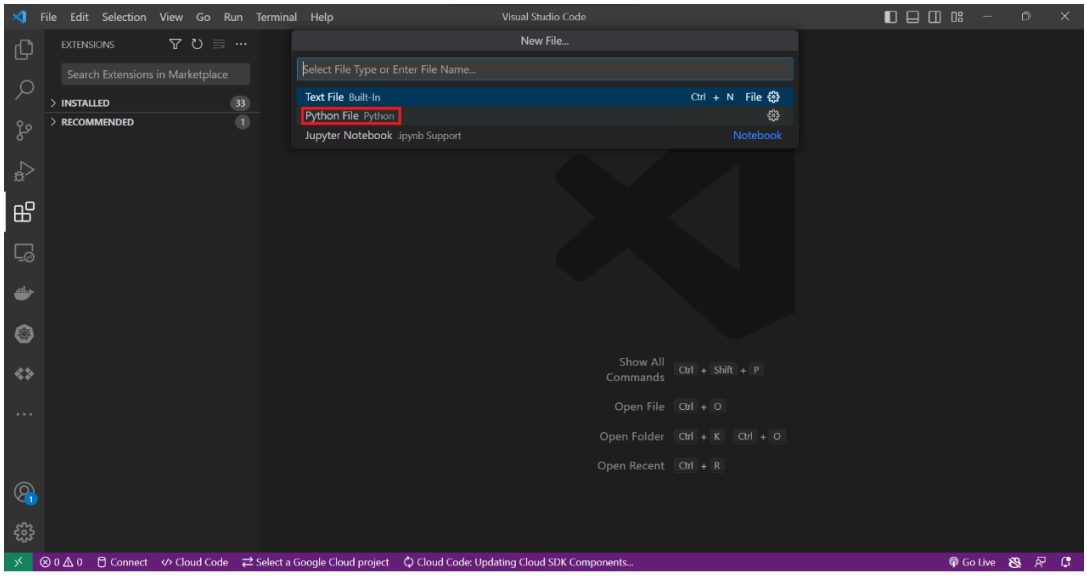


4. We will write a python code to load data from our local system. Search in windows Visual Studio Code and open it.

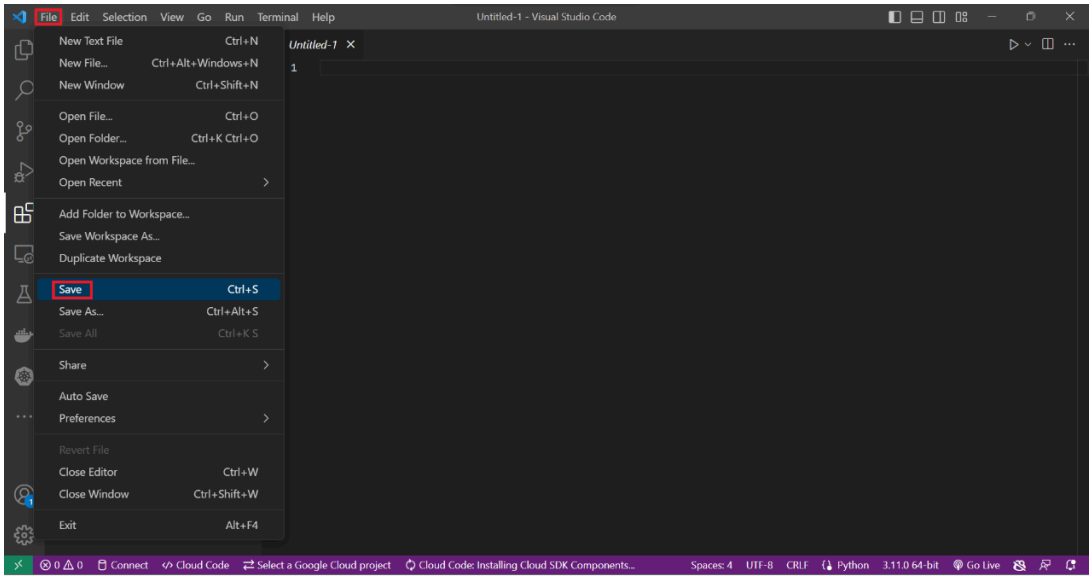
5. Click on File and then select New File option.



6. Select Python File and a new unnamed file will open.



7. Go to File and click on Save. Save this file to the directory that has your .csv files as we need to run the python from the same. Give it a name like ‘local_to_aws_s3.py’.



8. Write the below code to load files from your local system.

```
import boto3


# Connecting to AWS S3

s3_resource  = boto3.resource('s3', 'us-east-1') # remember to add your own region of aws s3


# Function to upload file in S3

def s3_upload(file_name,fold,bkt):
```

```
    s3_bucket = s3_resource.Bucket(name=bkt)


    if True:
        s3_bucket.upload_file(
            Filename = file_name,
            Key = fold + '/' + file_name
            )
        return True

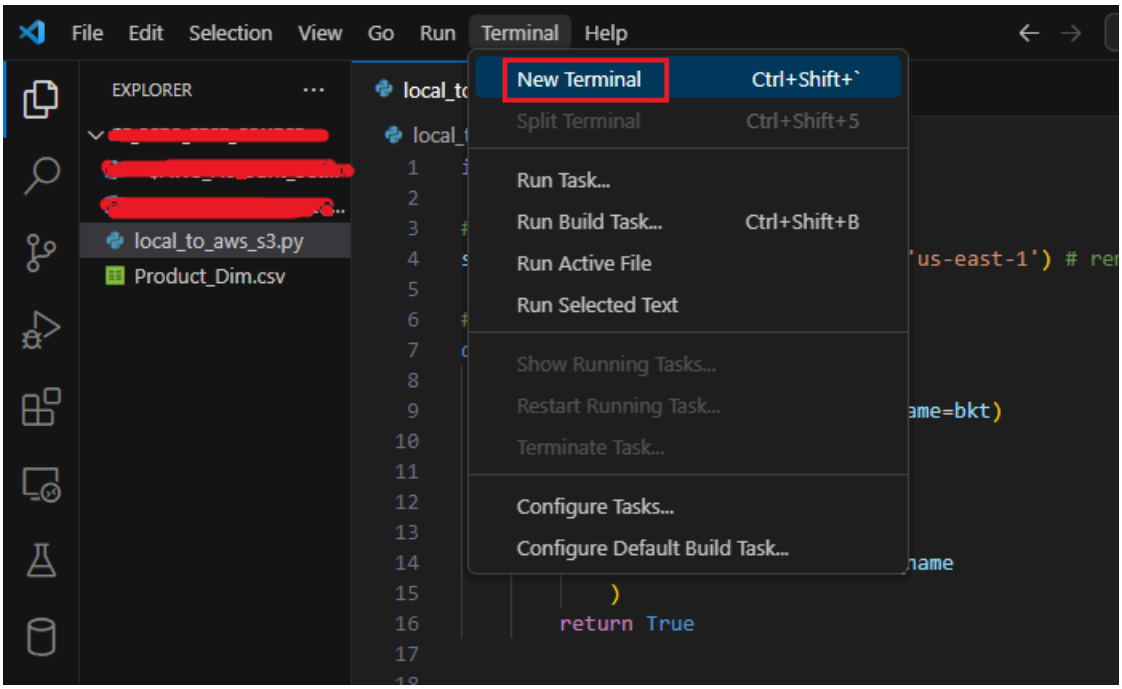

if __name__ == '__main__':


    file_name = 'Product_Dim.csv'
    s3_folder = 'raw_data'
    bucket = 'bucket-training-123'


    status = s3_upload(file_name, s3_folder, bucket)

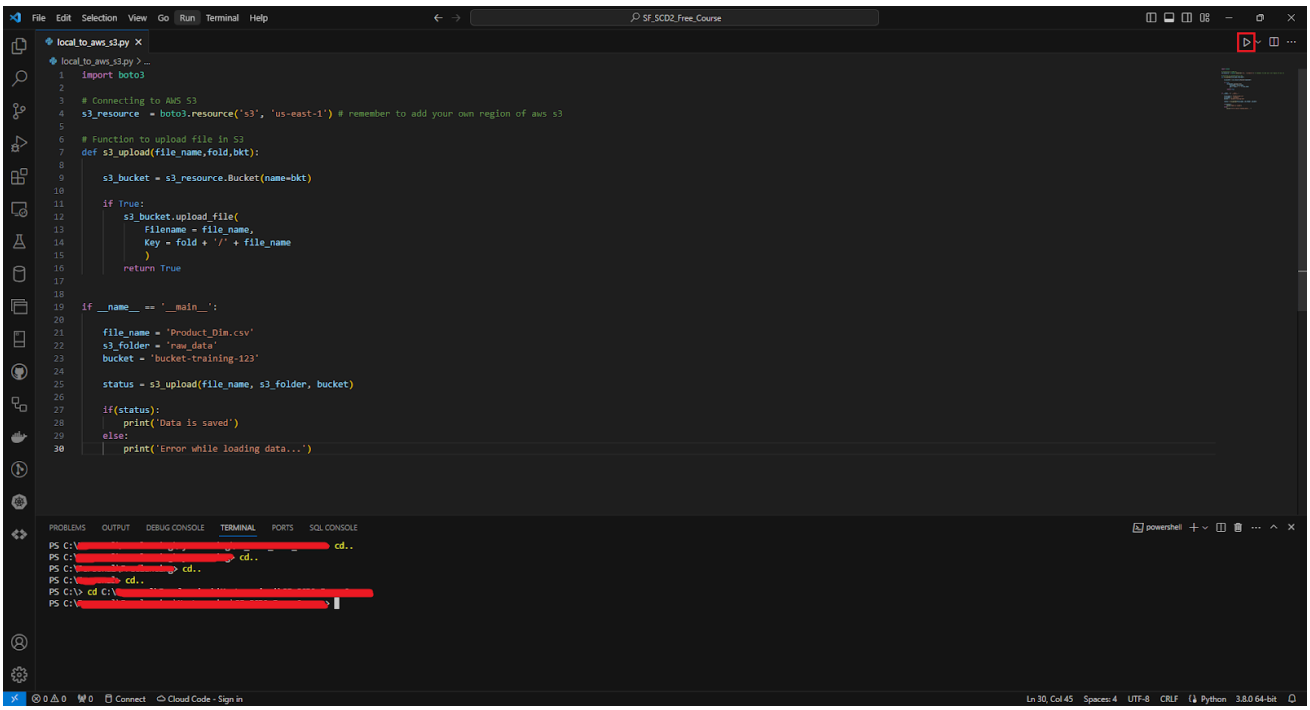

    if(status):
        print('Data is saved')
    else:
        print('Error while loading data...')
```

9. Now change your directory of the terminal in Visual Studio Code to your directory that has the files. Click on Terminal and select New Terminal.

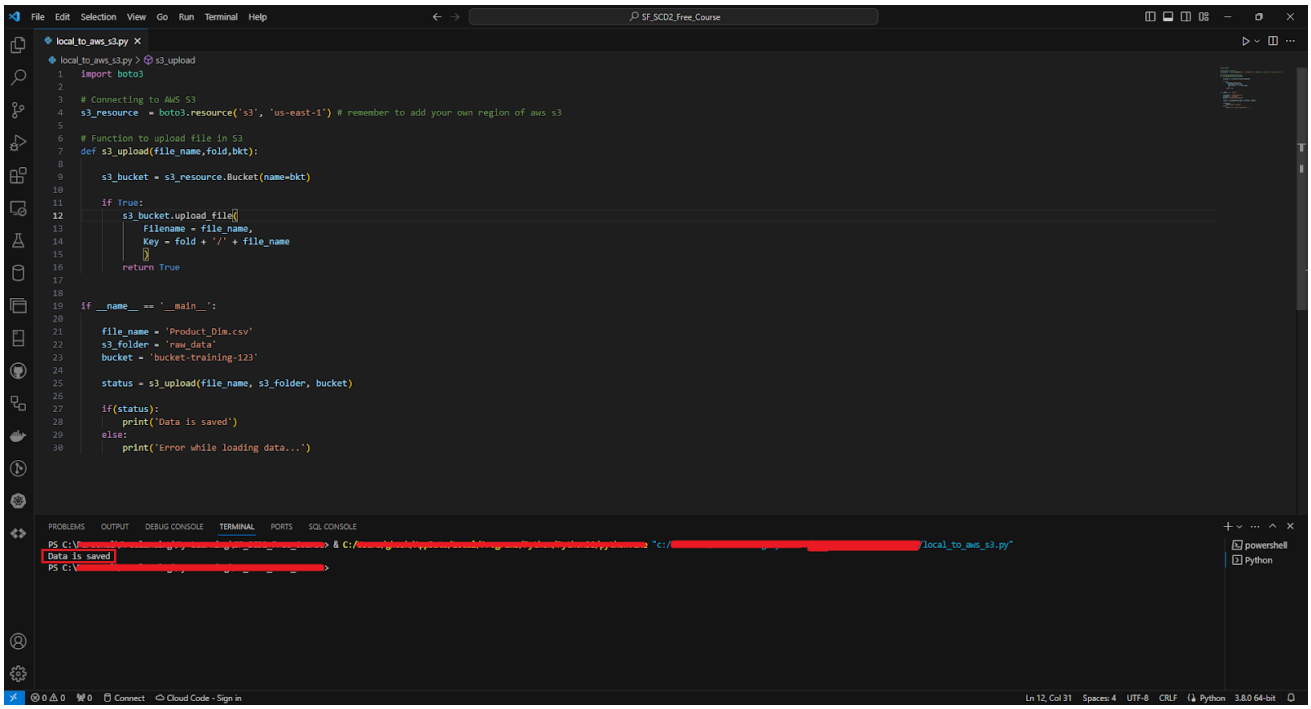


10. Run following commands one-by-one:

- cd.. [to get out of the system directory]
- cd <your directory>
- Then click on the Run button on top right corner, to run the file.



11. When the writing file to AWS S3 bucket folder is done then a message is displayed.



If you get and error, No module named ‘boto3’

pip install boto3

OR

python3 -m pip install boto3 (MacOS)

Successfully setup done of AWS CLI and AWS S3 bucket created. Data loaded using python script.

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